



Research Paper

Experimental assessment of different compact flow channel geometries on pressurised gas solar receivers

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ABSTRACT

Pressurised gas receivers using optimised compact flow channels exhibit promising thermal performance in concentrated solar thermal systems, paving the way to integrate solar energy in high efficiency thermodynamic power cycles and industrial heat processes. This work is focused on the experimental characterisation of several compact absorber samples, in order to verify previous numerical analysis aimed at maximising thermal efficiency while simultaneously minimising pressure drop. Four variants of plain rectangular channel absorber samples were fabricated varying channel height, breadth and wall thickness. Each absorber sample was experimentally studied by varying mass flow rate, inlet pressure and incident radiation flux.

The experimental campaign verified important findings and predictions of a previously developed numerical model including that the maximum thermal efficiency and pressure drop occurs at the smallest channel size besides the positive effect of taller and narrower channels. The maximum thermal efficiency observed was 95.8% with the corresponding pressure drop measured at 6.3% of the inlet pressure. This performance, in terms of thermal efficiency and relative pressure drops, is on par and even surpasses the state-of-the-art receivers of its type. Such high thermal efficiencies (above 90%) and low relative pressure drops (below 3%) were also observed for other operating conditions and absorber geometries as well.

1. Introduction

The solar thermal receiver is a critical component of any Concentrated Solar Power (CSP) or Concentrated Solar Thermal (CST) system as it serves to transfer the concentrated solar radiation incident on its surface as thermal energy to a heat transfer medium. The heat transfer medium employed in the receiver may be a solid, liquid or gas with this selection providing a means to classify receivers. Solids, especially as particles, are capable of attaining high temperatures and use as thermal storage media but particle conveyance, attrition and transport besides design difficulties in particle–fluid heat exchangers remain challenges for solid particle receivers [1,2]. Liquids, especially thermal oil and molten salt, are widely deployed and are the most prevalent Heat Transfer Fluid (HTF) in commercial systems [3,4] though the working temperature bounds of both these fluids create challenges. Thermal oils cannot operate above 400 °C due to decomposition and general chemical instability at higher temperatures [4]. This in turn precludes them from use in high temperature applications such as next generation

power systems, that typically require temperatures above 700 °C [5] or industrial processes [6]. Molten salts can achieve higher operational temperatures but have a high freezing point and solidify at around 240 °C which leads to salt crystallisation and the formation of plugs that creates blockages in the circuit [7]. Preventing this requires heat traces which is a major parasitic energy loss and their implementation incurs a significant material and financial cost [5]. Additionally, corrosion and erosion caused by molten salts, especially closer to the receiver when the molten salt becomes more chemically unstable at higher temperature, is a concern that needs to be monitored [8,9]. Nanofluids are also emerging as an active area of research and development in the CST field [10,11] but their application is limited to lower temperature systems such as parabolic trough collectors [12,13].

Gas phase receivers can obviate several of these constraints as gases are able to operate over a broad temperature and pressure ranges without physical or chemical degradation. Moreover, gases tend to not be toxic or environmentally hazardous while being relatively inexpensive [14]. On the downside, gases have unfavourable thermo-physical properties compared to liquids, which leads to poor heat transfer that

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Notation		L	Length (m)
<i>Acronyms</i>		$\dot{m}$	Mass flow rate (kg/s)
AISI	American Iron and Steel Institute	N	Number of channels/elements
CCD	Charged Coupled Device	Nu	Nusselt number
CST	Concentrated Solar Thermal	P	Pressure (Pa)
CSP	Concentrated Solar Power	ΔP	Pressure Drop (Pa)
EDM	Electrical Discharge Machining	Pr	Prandtl number
HCE	Heat Collector Element	$\dot{Q}$	Thermal power (W)
HTF	Heat Transfer Fluid	r	Radius (m)
HFSS	High Flux Solar Simulator	Re	Reynolds number
IR	Infrared	t	Channel wall thickness (m)
N ₂	Nitrogen (gas)	T	Temperature (K)
O ₂	Oxygen (gas)	v	Velocity (m/s)
PF	Perforated Fin	<i>Greek Letters</i>	
PFHE	Plate Fin Heat Exchanger	η	Efficiency
PRF	Plain Rectangular Fin	ρ	Density (kg m ⁻³)
sCO ₂	Supercritical Carbon Dioxide	Δ	Differential
SLM	Selective Laser Melting	ε	Roughness (m)
<i>Latin letters</i>		<i>Subscripts</i>	
A	Area (m ²)	abs	Absorbed
B	Breadth (m)	c	Channel, Contraction
D	Diameter (m)	e	Expansion
f _D	Darcy pressure friction loss factor	h	Hydraulic
f _F	Fanning pressure friction loss factor	inc	Incident
H	Height (m)	v	Vertical
K	Coefficient of pressure drop	Xe	Xenon

affects the thermal efficiency of the receiver and limits its operational temperature as the heat evacuation from the receiver solid itself is not as effective [15].

In this regard, the design of the flow channels in solar receivers plays a critical role as they must ensure sufficient heat transfer to the fluid without it undergoing excessive pressure drop. This is especially true when the heat transfer fluid is a gas as their compressibility lends them to additional pressure losses as compared to liquid media. These pressure drops may be alleviated by pressurising the system though this in turn adds to the system complexity and cost as additional appropriate pressure-rated components are necessitated [16,17]. The reduction in pressure drop is primarily due to a reduction in fluid velocity which occurs at the higher densities, corresponding to an increased pressure level, for the same mass flow rate [18].

Pressurised gas receivers can be grouped based on the gas employed (air, helium, nitrogen, etc.), the receiver irradiation pattern (directly, indirectly or hybrid) and the fluid flow path geometry [19]. The selection of gas used as the HTF in the receiver is a crucial system design parameter which may be done based on the application, optimal thermophysical properties of the gas, cost, etc [20]. In the early stages of pressurised gas receiver development, the primary application of said receivers was for pre-heating pressurised air before entering the combustion chamber in modified open gas turbine power cycles [21,22]. Given the application had pressurised air as the working fluid in the power cycle, the simplest system design called for the same pressurised air to be used as the HTF in the receiver. In other applications, such as industrial process heat, that usually employ an intermediary heat exchanger between the solar thermal loop and the process, alternative gases, with better thermophysical properties than air, may be used to improve the receiver and overall system efficiency [23,24]. More recently, the trend toward more thermally efficient supercritical carbon dioxide (sCO₂) power cycles [5,25] has lead to the investigation and development of pressurised gas receivers using sCO₂ as the HTF [26,27].

Classifying receivers based on irradiation pattern has them split into

directly irradiated receivers, which have their flow channels irradiated, and indirect receivers which have a solid physical separation between the irradiated surface and the flow channels of the HTF [19]. Directly irradiated receivers are limited in their operating pressure by the fragility of the quartz glass window used to allow incident light inside the pressurised receiver [28].

On the basis of the internal flow channel geometry, pressurised gas receivers may be classified, non-exclusively, as tubular, embedded channel, impinging jet and micro/compact channel [19]. Tubular receivers are by far the most studied and developed [22,29,30], though alternative concepts exist such as the embedded channel receiver [31], impinging jet receiver [32–34], Reticulated Porous Ceramic (RPC) lined cavity receivers [35–37] and compact channel receiver [38,39]. While tubular receivers boast low pressure drops of below 2 % of their inlet pressure, the thermal efficiency of such receivers does not exceed 80 %. Other proposed pressurised receivers have sought to improve on the performance of the conventional tubular receiver but inevitably face the same competing considerations of thermal efficiency maximisation and pressure drop minimisation. While some tubular receiver variants, notably the flat panel and bladed as well as embedded receivers, are able to achieve high efficiencies, they do so at the cost of excessive pressure drops well above 10 % of the inlet pressure. On the other hand, other receiver concepts, such as the RPC and impinging cavity receivers, have lower pressure drops below 1 % of their inlet pressure levels. However, the corresponding thermal efficiencies attained by these receivers is also lower, often below 60 %. Table 1 provides a summary of select pressurised gas receiver designs and performances, as well as the operating conditions under which they were experimentally characterised.

Compact flow channels offer several benefits such as greater heat transfer area for the same volume besides improving the heat transfer by convection due to decreased diffusion length as compared to conventional tubular channels [43]. The reduction in channel size results in a volume reduction for the same effective heat transfer area, i.e. its compactness, which in turn causes enhanced heat transfer coefficients

Table 1

Pressurised gas receivers: state of the art with performance and operating conditions.

Receiver	Operating conditions		Inlet Pressure bar	Performance indicators		Ref
	Inlet Temperature °C	Outlet Temperature °C		Thermal efficiency %	Pressure drop mbar %	
SOLGATE	290	350–550	6.5	75	100–120 1.5	[21]
SOLHYCO	600	749–783	3.75	38–45	67–72 1.87	[22]
SOLUGAS	300–350	800	9	71.3–78.1	200 2.2	[30]
Embedded circumferential	135	520	3.81–3.9	83–96	1570–1940 46	[31]
Embedded axial	150	450	3.85–4.05	78–96	410–550 12	[31]
Impinging jet	674	800	3	74.1	– –	[40]
Microchannel (CNRS)	$\Delta T = 100 - 360$		4	64.28	750 18.75	[39]
Sunshot (Brayton)	150	450	250	90.6	< 7500 < 3	[41]
Microchannel (OSU)	400	650	81–86	90	< 4000 < 2	[42]

for the same mass flow rate. However, a higher pressure drop will also occur which could be offset by pressurisation.

Reviewing the current state of the art in the study of such compact channel pressurised gas receivers, specifically focussing on experimental works, a microchannel receiver prototype was developed [38,39] based on compact heat exchanger concept. It was divided into 12 parallel channels, each being 1 mm wide and 3 mm high. Rectangular ribs were attached on top of each channel, to improve the heat transfer to fluid, having the same width as the channel and a height and pitch of 2 mm and 1 mm, respectively. The receiver was manufactured using Selective Laser Melting (SLM) in Inconel 625. The outer (or top) surface of 30 mm $\times$ 30 mm was the irradiated plane. An experimental campaign was carried out using pressurised air (2–6 bar) as the HTF and thermal efficiency of around 64 % was observed with a pressure drop of around 750 mbar.

On a smaller scale, absorber panels with a Plated Fin Heat Exchanger (PFHE) type structure and channel sizes in the order of hundreds of micrometres were manufactured of Haynes 230 using Electrical Discharge Machining (EDM) [42,44]. Lab-scale absorber panels of 2 cm $\times$ 2 cm were fabricated and demonstrated to absorb 100 W cm⁻² of incident flux using supercritical carbon dioxide (sCO₂) at 650 °C and 80–200 bar as the working fluid, reporting thermal efficiencies around 90 %.

Another receiver prototype, using sCO₂ as the HTF and internal flow channels based on PFHE concepts, was developed by the National Renewable Energy Laboratory (NREL). The receiver consisted of a compact structure with two attached plates having a wavy fin arrangement between them. The plates, which may be stacked one atop the other, act as the absorber surfaces of the concentrated solar radiation and are arranged to form a cavity [41]. To withstand the higher internal pressures demanded of a sCO₂ receiver (up to 25 MPa) while simultaneously bettering heat transfer to fluid, a high fin density (up to 32 fins/cm) and thin walls were designed. The receiver, and its individual panels, were constructed of Inconel 625 and was predicted to have an efficiency of around 90 %.

From the review of past studies into pressurised gas receiver design and development, it is evident that an overarching objective was to maximise heat transfer to the gas, and ultimately the receiver thermal efficiency, without suffering excessive pressure drops in the process. To this end, in a previous work, the authors of this manuscript presented a numerical analysis into the comparative performances of pressurised gas receivers with various compact flow channel geometries which concluded that the Plain Rectangular Fin (PRF) and Perforated Fin (PF) channel receivers performed best in terms of exergy and energy efficiencies [45]. Furthermore, the influence geometrical parameters, common to all evaluated receiver flow channel geometry types, was explored and uncovered optimum geometric configurations that maximised the exergy and energy efficiency respectively. The four common geometric parameters selected were the channel breadth, height and wall thickness besides the number of vertical channels (rows). The exergy efficiency, which accounts for heat transfer to HTF and its

pressure drop, was largest in receivers having taller and narrower channels with thicker walls. The energy efficiency, which only depends on the solar to HTF heat transfer, was maximised in receivers having the smallest channels with the fewest number of vertical channels. As the exergy efficiency also factored in the pressure drop, it was found to be a valuable performance indicator of the receiver as excessive pressure drops are an inherent, though undesirable, feature of gas receivers. The parametric analysis concluded that there were optimum geometrical configurations that maximised the exergy efficiency and an experimental study was proposed to verify these findings.

As a natural continuation of the previous numerical analysis of pressurised gas receivers, this work presents the experimental evaluation of pressurised gas receivers based on the same compact channel geometries studied in the numerical models. More specifically, this work focuses on the experimental investigation into the effects of receiver flow channel geometry, besides other operational parameters, and its comparison with the previously developed numerical model. Through this experimental campaign, this work seeks to verify the predictions of the pressurised gas receiver numerical model and demonstrate the potential of pressurised gas receivers with optimised compact flow channel geometries for concentrated solar thermal applications. Attaining high thermal efficiencies (above 90 %) at low pressure drops (below 5 % of inlet pressure) is the perennial challenge of solar receiver design, which most state-of-the-art receivers fall short of as shown in Table 1, and in this experimental work, it is attempted to achieve these performance objectives and analyse the factors affecting the proposed receivers' performances and in so doing contribute to the field of solar receiver development.

The structure of the manuscript is as follows. The experiment design and preparation are described in Section 2. Firstly, the experimental setup itself, including the test facility, its operational boundaries and instrumentation is presented in Section 2.1 followed by a description of the absorbers designed and manufactured for experimentation in Section 2.2. Multiple absorbers, each with different internal geometries, were tested in this campaign. The experiment design and overall plan of experimentation is laid out in Section 2.3. This section details the operating parameters chosen to be varied and the levels of each parameter to carry out a characterisation of each fabricated absorber to evaluate the performance of the absorber at each operating point. Section 3 delves into the results of the experiment campaign. The performance of each of the absorbers tested is presented and compared against one another as well as to the numerical model of the pressurised gas receiver. Finally, Section 4 presents the conclusions of the work as well as propose future lines of research.

2. Experiment design and preparation

2.1. Experimental facilities and operation capacities

The experiments were performed using a specifically designed pressurised receiver testbench making use of the KIRAN-42 High Flux

Solar Simulator (HFSS) [46,47] as a concentrated sun-like light source, at the IMDEA Energy, Madrid (Spain). The KIRAN-42 HFSS acted as the primary radiative source and is comprised of 7 independent xenon short-arc lamps, each of 6 kW (electrical), with truncated ellipsoidal reflectors. The radiation flux profile on the target plane can be altered by varying the number of operational lamps, the pointing of individual lamps and the position of the target plane relative to the HFSS [48,49]. A calorimetric testbench is required to characterise and analyse the performance of the pressurised gas absorbers by quantifying the heat transfer from the incident solar thermal radiation to the HTF. Besides this, the testbench is also required to provide some level of control over key operating parameters to permit an experimental characterisation of the absorber. The pressurised gas receiver testbench facing the KIRAN-42 HFSS is depicted in Fig. 1.

The experimental facility was designed to be inherently flexible in terms of operating conditions. Multiple receiver concepts may be tested as individual absorbers or in some combination of the same (series/parallel) given the multi-lamp KIRAN-42 HFSS and its manipulable pointing strategies. Furthermore, aspects related to the absorber itself including the material used, external geometry, internal flow channel geometry and surface treatment were variable within limits of the testbench and desired experimental conditions though this will be discussed in more detail in Section 2.2. Operating parameters related to the calorimetric testbench such as the gas used, pressure level, mass flow rate and inlet temperature could also be varied and controlled within the constraints of the specific instrumentation used. The process and instrumentation diagram of the experiment facility is presented in Fig. 2. Referring to Fig. 2, when starting from the pressurised air cylinder, the high-pressure air passes through a pressure regulator, that delivers a constant output pressure as required by the experiment, before passing through a mass flowmeter. Compressed air cylinders with a fixed composition of 79 % Nitrogen (N_2) and 21 % Oxygen (O_2) were used to supply the pressurised air throughout the experiment campaign. Absolute pressure sensors are placed before and after the absorber besides thermocouples to quantify the heat transfer to the HTF as the properties of the gas vary as a function of its pressure and temperature. This experiment data was inputted into the fluid property database REFPROP [50] to obtain the thermophysical properties at the absorber inlet and outlet with which the performance could be evaluated. A series of thermocouples are also proposed at various locations on the surface of the absorber to assess the solid temperature and its variations across the length and breadth of the absorber. The heated air then passes through a

length of coiled tubing submerged in a cooled water tank before the air is eventually discharged to the environment at room temperature via a needle valve to ensure adequate backpressure.

The system instrumentation also includes a charged-coupled device (CCD) camera and a thermographic or infra-red (IR) camera. Both these cameras are situated behind the lamps of the KIRAN-42 HFSS and are pointed towards the testbench. The CCD camera, with a light intensity filter, is used to safely view the testbench from the front during lamp operation and is essential in generating the radiation flux profiles/maps. The thermographic camera allows analysing infrared radiation from the absorber. A Gardon-type heat flux sensor is utilised to quantify the radiation flux incident on the absorber. Table 2 details some important instrumentation used in the experimental setup including its operating range and instrument error. The experimental test setup, along with its instrumentation, can be observed in Fig. 3. Further information about the experimental setup can be found in [51] which includes photos and annotated drawings of the key components, their relative positions on the testbench and their functions.

Data from the instrumentation (excluding cameras) was communicated through National Instrument data acquisition devices, including C series modules and chassis, to the LabVIEW software [52]. Through LabVIEW, experiment monitoring and control was executed and experiment data was saved at regular time intervals. This data was then processed in MATLAB [53] to analyse the experimental data and assess absorber performance.

2.2. Absorber design and fabrication

As the absorbers to be tested were of a relatively small size and were designed for a one-off experiment campaign, additive manufacturing was identified as the preferred means of fabrication. Absorbers of stainless steel (316L) manufactured by Selective Laser Melting (SLM) will be presented in this work with this material selection guided by ease of manufacturability and its, relatively, high temperature strength. The pressurised receiver numerical model indicated that the perforated fin and plain rectangular fin receivers have the best performances [45]. Given the slight but inescapable inaccuracies of 3D printing holes and overhangs (sags and shape deformations) [54], it was decided to manufacture all the absorber samples using the plain rectangular fin compact flow geometry. Absorbers of different internal flow geometries were proposed to be built to investigate the effect of the variation of these geometrical parameters on the performance. The four geometrical

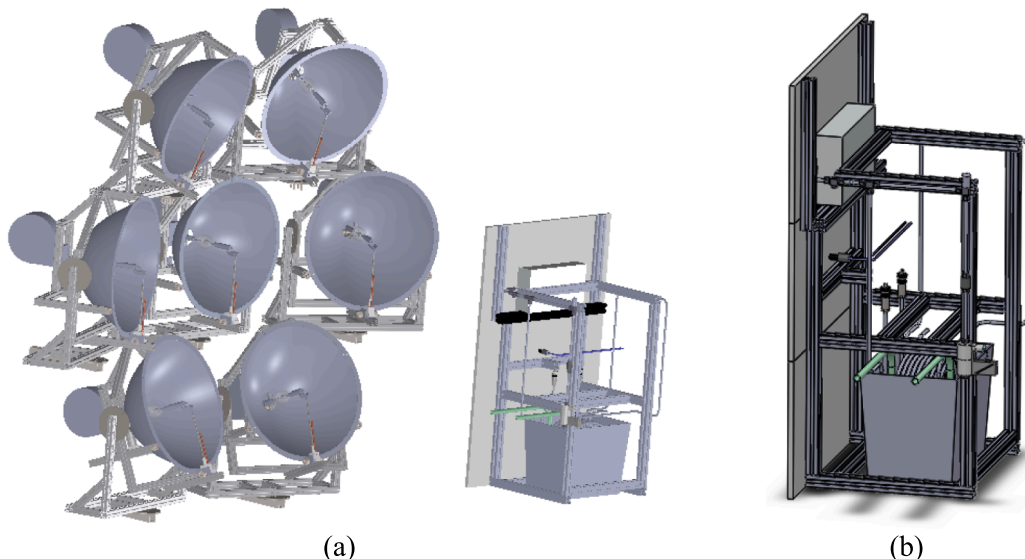


Fig. 1. Experimental setup developed for gas phase receiver testing with KIRAN 42 HFSS; (a) entire setup as in position, (b) testbench close-up projected view.

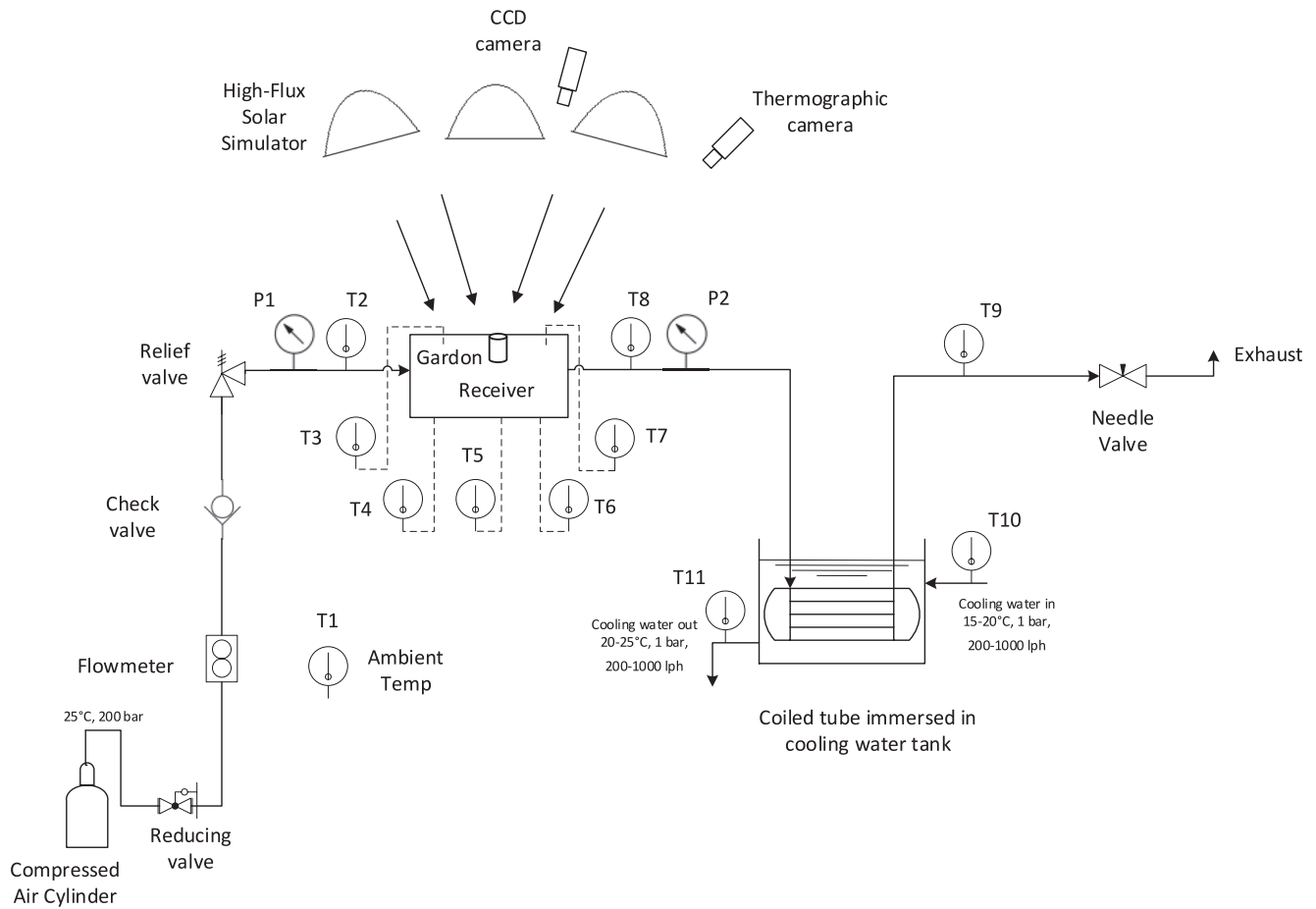


Fig. 2. Process and instrumentation diagram of experimental test setup.

Table 2

Instrumentation used in experimental setup.

Instrument	Manufacturer	Model	Measured parameter	Unit	Operating / Measurement Range	Instrument Error
Gardon flux gauge	Vatell Corporation	TG-1000	Radiation flux	kW/m^2	0 – 1600	3 %
Thermocouples	TC Ltd	Type K (class 1)	Temperature	$^{\circ}\text{C}$	Continuous: 0 – 1100 Short Term: –180 – 1350	$\begin{cases} \pm 1.5, -40 < T < 375 \\ 0.4\%T, 375 < T < 1000 \end{cases}$
Mass flowmeter	Bronkhorst	F-111AC	Mass flow rate	g/s	0 – 2.15	0.5 %
Pressure transducer	TC Ltd	716-912	Absolute pressure	bar	0 – 16	0.5 %
Spectrophotometer	PerkinElmer	LAMBDA 1050+	Radiation absorptivity	–	175–3300 nm	1 %

design parameters identified and varied in the numerical model, namely channel height (H_c), wall thickness (t), number of vertical channels (N_{cv}) and channel breadth (B_c), were also varied in the experimental absorbers. As there are four geometrical parameters varied, there will be four absorber variations, as laid out in Table 3, labelled PR1, PR2, PR3 and PR4.

As can be observed in Table 3, PR1 has square channels whilst PR2 and PR3 have taller and broader channels respectively. PR4 has a thinner channel wall because of which an additional channel, along the horizontal row, is added. Fig. 4 provides a better visualisation of the differences in each absorber's cross-section. The design of the absorbers had to factor in thermal, mechanical and manufacturing considerations. Numerical analysis [45] revealed that the optimum velocity, for good heat transfer at lower pressure drops, was around 20–30 m/s while the additive manufacturing method, given that the internal flow channels would be challenging to access and post-process, recommended minimum channel sizes of 1 mm. The wall thicknesses were determined using the design by rule approach following the ASME Section VIII Division I [55].

Each of these absorbers have been designed to have the same outer dimensions, to facilitate mounting and dismounting besides to simplify the insulation design. The absorber outer dimensions were reached after consideration of the operational limitations of the experimental facilities, including the KIRAN-42 HFSS and air supply, resulting in an absorber length of 63.75 mm, breadth of 24 mm and height (depth) of 11 mm. Additional information regarding the design of the absorbers can be found in [51]. Once the design of the four absorbers was established, numerical modelling and simulation was performed to verify that the maximum temperature and temperature gradients were within limits. Identical inlet and outlet sections were designed on either end of the absorber and manufactured as one piece as is evident in Fig. 5(a). Only one face of the absorbers was exposed to the incident radiation with the remaining faces insulated as shown in Fig. 5(b).

2.3. Experiment design and method

In order to conduct an experimental characterisation of the four absorbers, three operational parameters, namely peak incident flux,



Fig. 3. Pressurised receiver test setup as seen from front (left) and back (right). Absorber centrally located in rectangular aperture in radiation shield facing frontward.

Table 3
Internal flow channel geometric parameters of 4 experimental absorbers.

Receiver (notation)	Geometric Parameters			
	H_c (mm)	t (mm)	N_{cy} (rows)	B_c (mm)
Plain Rectangular (PR1)	1	3	2	1
Plain Rectangular (PR2)	2	3	2	1
Plain Rectangular (PR3)	1	3	2	2
Plain Rectangular (PR4)	2	2	2	1

mass flow rate and air inlet pressure, were varied at the levels/points given in Table 4. The upper and lower limits of these operational parameters were set based on practical design considerations of the test-bench and absorber. For example, the upper limit of the mass flow rate was calculated based on the expected experiment runtime and pressurised gas cylinder capacity so that an experiment run could always be performed using a single cylinder. On the other hand, the lower limit of the mass flow rate and the upper limit of the peak incident flux were set by numerically evaluating the risk of overheating the absorbers and preventing the same by ensuring sufficient gas velocity in the flow channels and controlled incident heat. The minimum peak flux limit was determined as that needed to achieve a significant temperature rise in the gas. A more detailed explanation of the experiment design, including the use of a previously developed numerical model [45] and learnings from preliminary trials, can be found in [51]. The inlet pressures were controlled to be within limits of those used in similar experimental studies [39] as it is understood that, while pressurisation reduces pressure drop in the absorber, higher system pressure levels require more complex and expensive piping, fittings and instrumentation with a suitable pressure rating that offsets performance gains in reduced pressure drops.

Thus, for each absorber, 24 experiments were conducted at each

combination of the operational parameters according to the experiment procedures laid out in the Appendix and [51] which includes supplementary data related to each experiment run and the design of experiments. Briefly, a typical experiment run entailed maintaining a fixed mass flow rate and inlet pressure and gradually increasing the incident radiation flux. At each incident flux level, the absorber was allowed to reach 'steady-state' or thermal equilibrium which was taken as when the air outlet temperature stabilised to within 1 % of its mean value. This normally took around 7–8 min though longer wait times were given to ensure that the absorber was stable and to allow for 5 min for the recorded or measurement interval. An experiment run can be observed in Fig. 6 in which the mass flow rate of air is kept constant while the incident flux level increases four times which causes the outlet temperature to also rise to four distinct levels.

It was initially proposed to also operate at 8 bar but this operating inlet pressure point had to be scrapped due to time constraints. The incident flux profiles corresponding to each of the four peak incident flux levels listed in Table 4, are presented in Fig. 7. The characterisation of the KIRAN-42 HFSS, and the generation of the radiation flux maps, was performed before testing with the absorber using the indirect characterisation or flux mapping method [49] that requires the use of a Gardon heat flux sensor and CCD camera. A more detailed elaboration of the characterisation technique used can be found in [51] which also presents the procedures and guidelines for radiation flux mapping.

In addition to the radiation flux profiles, Table 5 presents the values of radiation on the absorbers, as well as their distribution, which has been used and will be presented in the subsequent performance analysis.

The performance indicators used in this study are the pressure drop and the energy and thermal efficiencies which are defined in Eqs. (1) and (2) respectively.

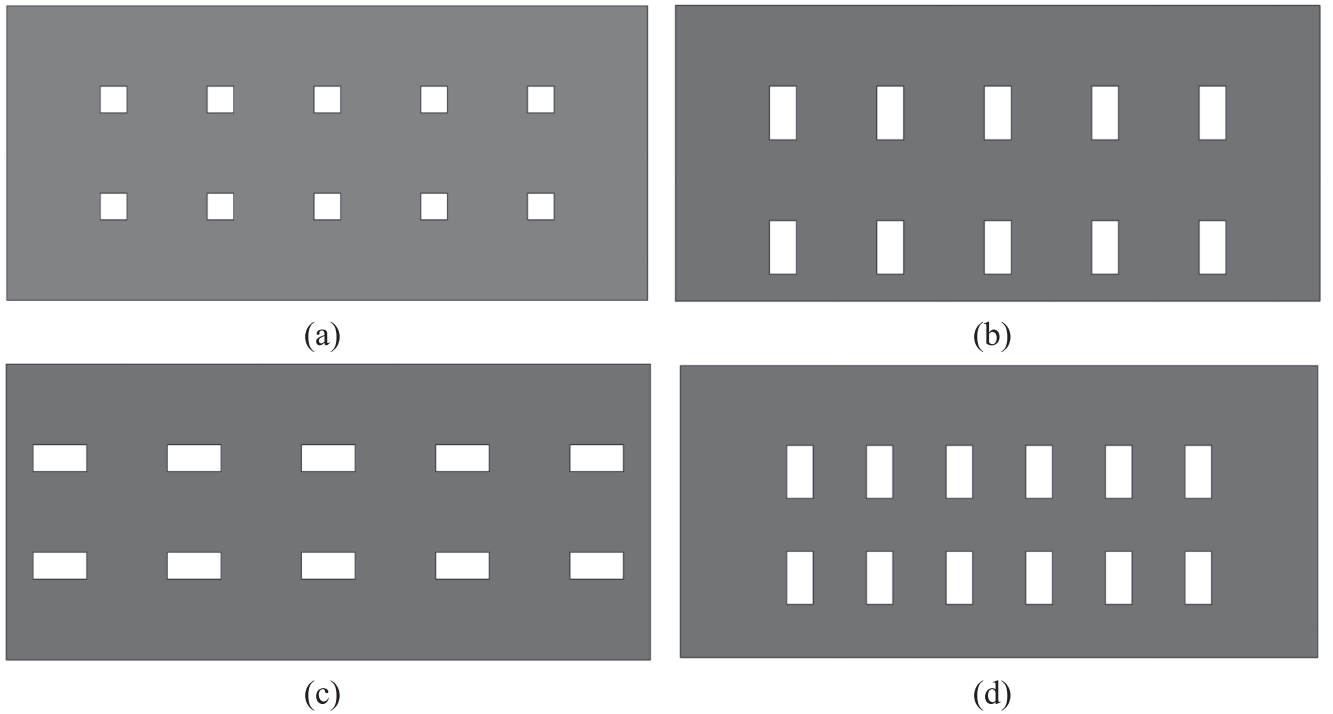


Fig. 4. Cross-sectional view of experimental absorbers: (a) PR1, (b) PR2, (c) PR3, (d) PR4. Flow channels in white. Absorber internal dimensions as in Table 3.



Fig. 5. Experimental absorber: (a) As manufactured, (b) In position on testbed with insulation.

Table 4
Operating control parameters and respective datapoints.

Parameter	Unit	Limits	Operation set-points
Peak Incident flux	kW/m ²	200 – 2000	200 – 300 – 400 – 500
Mass flow rate	g/s	1 – 2	1 – 1.50 – 2
Inlet Pressure	bar	3 – 15	4 – 12

$$\eta_{\text{energy}} = \frac{\dot{Q}_{\text{HTF}}}{\dot{Q}_{\text{Xe,inc}}} \quad (1)$$

$$\eta_{\text{thermal}} = \frac{\dot{Q}_{\text{HTF}}}{\dot{Q}_{\text{abs}}} \quad (2)$$

where $\dot{Q}_{\text{HTF}}$ is the useful heat transferred to the HTF, $\dot{Q}_{\text{Xe,inc}}$ is the total incident xenon radiation on the receiver and $\dot{Q}_{\text{abs}}$ is the absorbed radiation. Exergy efficiency was used when analysing the compact flow channel geometry variants in the preceding numerical analysis [45] and was found to be a useful tool for the optimisation and comparison of receiver types. However, this function cannot be directly employed in

this experimental study due to practical considerations. Primarily, the exergy of the incident radiation on the absorber from the multi-lamp HFSS, considering the optics of the HFSS and the short-arc xenon light source, has not been evaluated and is, as such, beyond the scope of this work. Moreover, the exergy gain [56], or alternatively entropy generation [57,58], in the HTF has to be evaluated at multiple intervals given the significant change in the HTF properties especially along the absorber length. While this is easily achieved in the numerical model by discretisation of the absorber into multiple Heat Collector Elements (HCEs), the current experimental setup only has thermocouples and pressure sensors before and after the absorber. This provides an aggregate representation of the absorber performance but it fails to capture the internal phenomena within the flow channels along the flow length and thus the exergy gain cannot be accurately gauged.

3. Results and discussion

3.1. Observations during experimentation and subsequent analyses

The four absorbers were individually experimented upon, at each of the planned operating points, and the experimental observations were

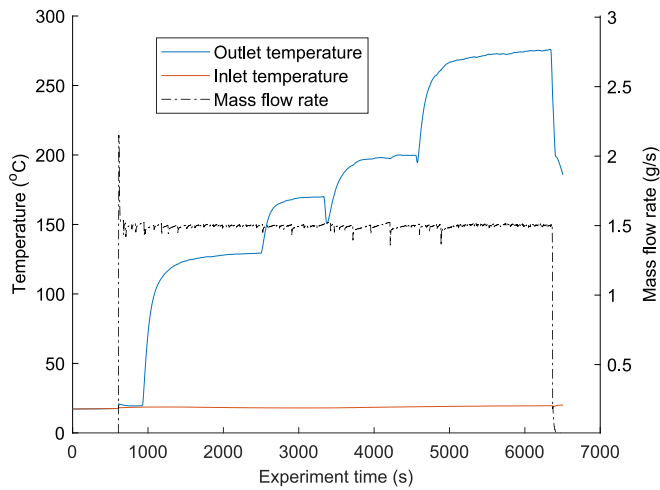


Fig. 6. Typical experiment run with single absorber at constant mass flow rate and increasing outlet temperatures corresponding to increasing incident flux levels.

compared with the pressurised receiver numerical model. This numerical model had to be modified to introduce some of the experimental conditions such as non-uniform incident radiation flux, roughness in the

absorber channels, surface absorptivity and pressure drops that were previously not factored into the model. These modifications are detailed in the [Appendix](#).

Over the course of the experiments, it was observed that prolonged operation at high temperatures led to oxidation of the absorber which serves to discolour (darken) it as is evident in [Fig. 8](#) where the absorber is uniformly coloured before and non-uniformly discoloured post-experimentation.

Analysing this oxidation was further complicated by the fact that it was a continuous process depending on the local temperature on the absorber walls. Consequently, the surface absorptance was not uniform throughout the experiment run of the absorber but continuously changed. The absorptance was measured before the start of experimentation on the absorber and subsequently only after all the

Table 5

Incident radiation at the peak flux levels used in the experiment campaign.

Peak radiation flux (kW/m ²)	Mean radiation flux (kW/m ²)	Standard deviation (%)	Radiation on absorber (W)
200	161.8	6.8	252.9
300	232.5	1.4	370.3
400	273.0	2.1	434.9
500	388.4	2.9	611.5

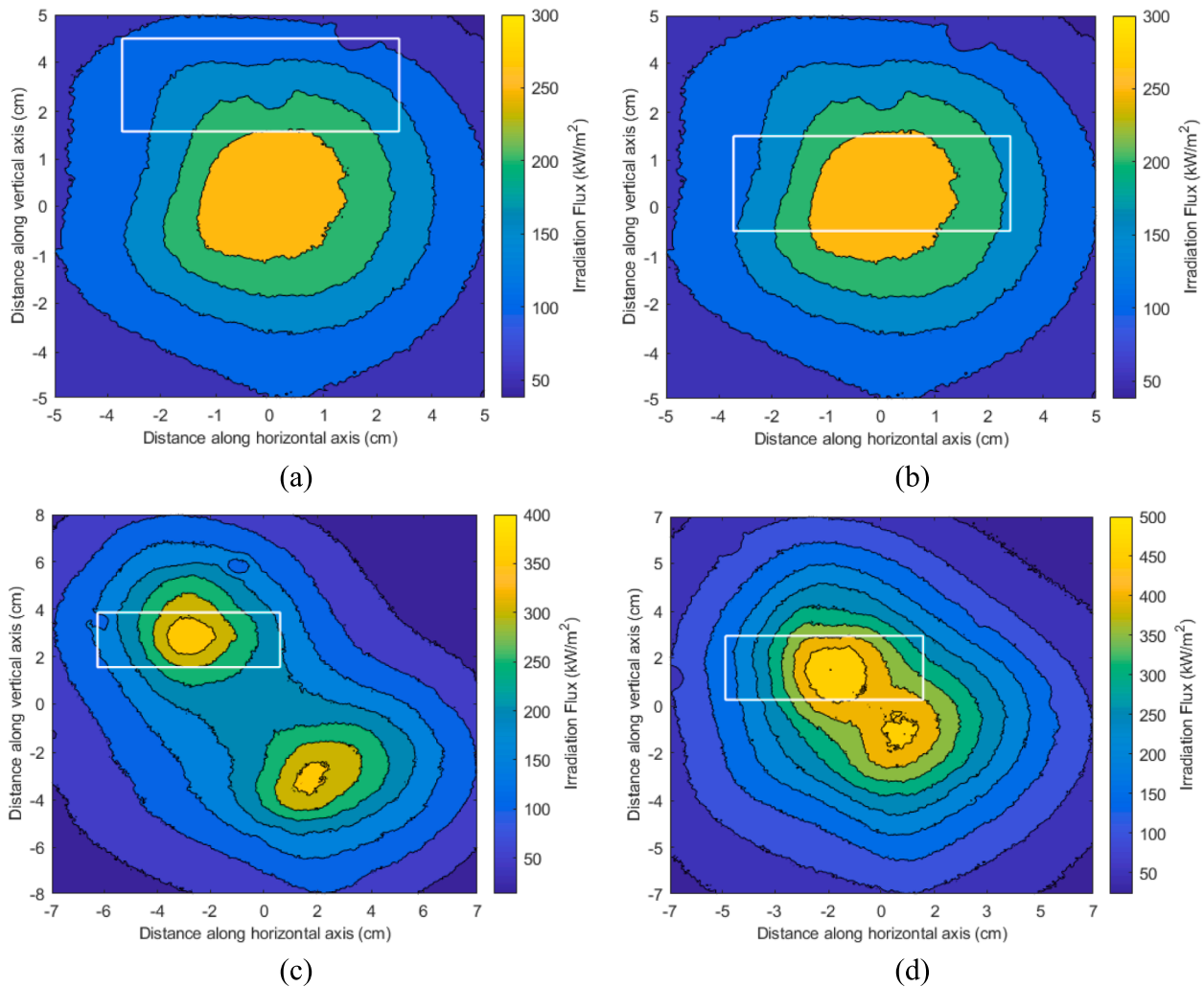


Fig. 7. Incident radiation profile at different peak flux levels used in experimental campaign: (a) 200 kW/m², (b) 300 kW/m², (c) 400 kW/m², (d) 500 kW/m². White rectangle represents the incident exposed surface of absorber.

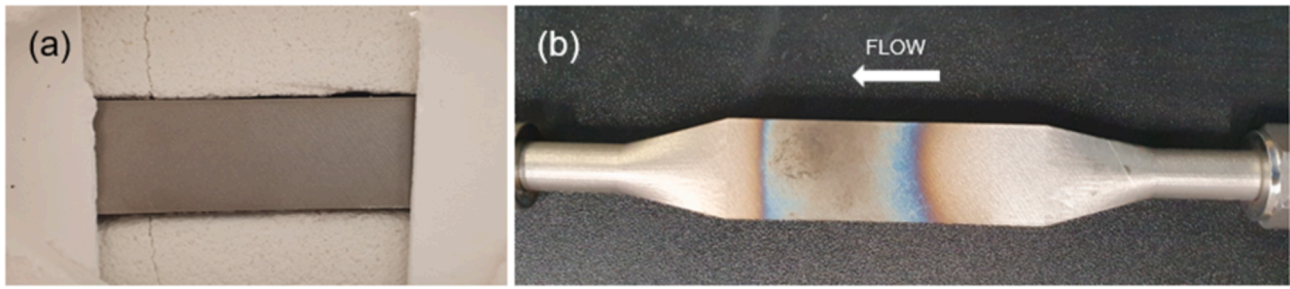


Fig. 8. Absorber front: (a) before experiments, (b) after experiments. In the photographs, the absorber inlet and outlet are at the right and left ports, respectively.

experiments on each absorber were conducted. This was due to the infeasibility of removing the absorber, testing it in a spectrophotometer UV-Vis Lambda 1050 (Perkin Elmer) and remounting it in the testbed for each experiment operation point.

To overcome this challenge, multiple photos of the absorbers were recorded and their image processing, using the measured absorbance data and corresponding images as reference, was performed to estimate the absorbance before and after each experiment run. The aggregate results of this analysis can be observed in Fig. 9, though it should be noted that for the subsequent analysis, each absorber was subdivided into three regions (inlet, centre and outlet) with different absorbances as laid out in the Appendix. Fig. 9 is solely intended to demonstrate the trend of rising absorbances as the experiments wore on and general utility of the image processing technique employed to estimate the absorbances between the initial and final measured values.

For intermediate experiment runs, the absorber was not removed and instead the absorbance was estimated using image based techniques, interpolating between the known absorbance measurement points determined with the spectrophotometer. When using one of the measurement points as reference, this image based technique, developed as a MATLAB routine using its Image Processing Toolbox, was able to predict the other point's absorbance within 5 % of its reading with the spectrophotometer. It was hence deemed a practical means with which to determine the absorbance values of the absorbers in the intermediate experiment runs.

An experiment run typically consisted of multiple experiment operation points increasing the incident radiation flux while keeping the mass flow rate and inlet pressure constant as indicated in the operation procedure which is detailed, along with ancillary test and precautionary procedures, in [51]. The xenon weighted absorbances of the absorbers before and after the entire set of experiments was, on average, around

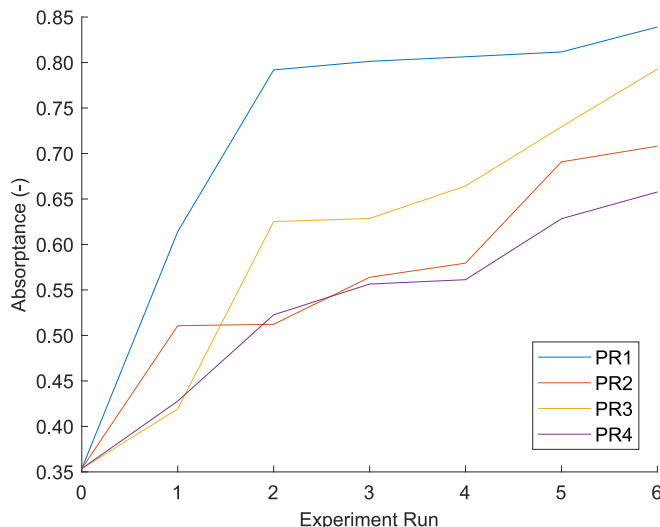


Fig. 9. Average absorbance after each experiment run.

0.35 and 0.75 respectively.

The absorbers were relatively small and metallic and hence have low thermal inertia. Thermal equilibrium was achieved within a few minutes and could hence be described as thermally stable at any given operating point. As a precautionary measure during experimentation, it was decided to abort a given operating point if the temperature of the absorber solid rose above a safe threshold. This primarily occurred at high incident radiation fluxes and low mass flow rates. Nevertheless, the gradual temperature rise allowed sufficient time to implement mitigating solutions such as defocussing the solar simulator or increasing the mass flow rate and hence even in extreme conditions the system performed as expected and was simple to control.

3.2. Experimental characterisation of individual absorbers

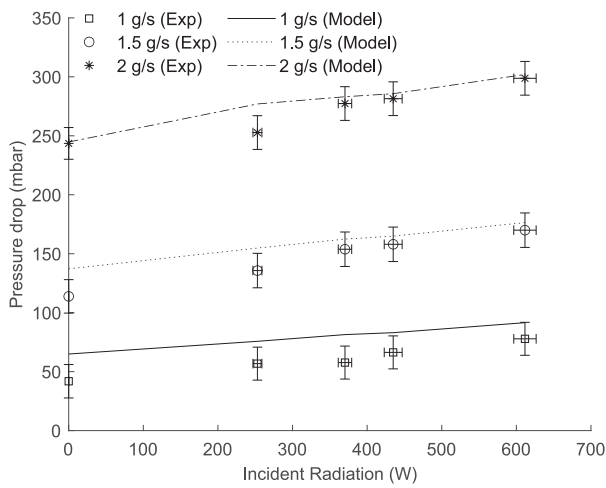
Though all four absorbers were experimentally characterised, this section will focus on the results of the PR1 absorber before a comparative analysis is made with the other absorbers in the subsequent section. This is simply to limit the quantum of graphical information which is otherwise similar, in behaviour, among all four absorbers regarding performance patterns with varying input parameters. The individual absorber plots and details may be found in [51]. The pressure drop variations with incident radiation flux and mass flow rates are presented in Fig. 10. It should be noted that the incident radiation levels plotted on the x-axis correspond to the peak flux values listed in Table 4 with the zero value indicating a 'cold' reading before the lamps are switched on.

Observing Fig. 10, the pressure drop rises with increasing incident radiation levels. This is because of the increase in the channel velocity with higher fluid temperatures, and hence lower densities, expected when the incident radiation level rises. Higher channel velocities cause higher pressure drops. On the other hand, at greater inlet pressure levels, seen in Fig. 10(b), the pressure drop falls which can be attributed to the larger density at elevated pressures and thus the smaller channel velocity resulting in a lower pressure drop. Higher mass flow rates cause greater channel velocities, if the cross-sectional flow area is constant, and this explains the increased pressure drops at higher flow rates. The numerical model generally lines up reasonably well with the experimental data. The pressure drops coincide well given that the exact value of surface roughness, which plays an important role in the pressure drop, is unknown and an estimated value, based on relevant studies into the surface roughness during similar manufacturing techniques, has been employed [59,60].

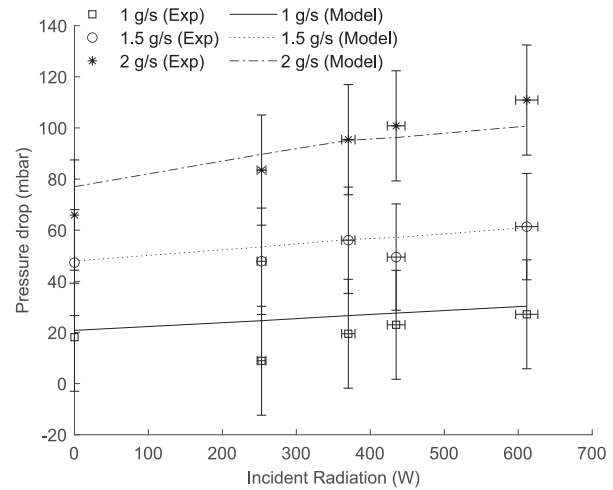
To further evaluate the effect of this surface roughness in dimensionless terms, the friction factor was calculated from the experimentally measured pressure drop using Eq. (3):

$$f_D = \frac{2 \Delta P D_h}{\rho_c v_c^2 L_{abs}}, \quad (3)$$

where f_D is the Darcy friction factor, ΔP is the pressure drop, D_h refers to the hydraulic diameter, v_c is the velocity of fluid in the channel and L_{abs} is the length of the absorber. The channel velocity is calculated after first measuring the average density of the air, using the absorber inlet and



(a)



(b)

Fig. 10. Experimental results of absorber PR1 compared with numerical model. (a) Pressure drop at inlet pressure of 4 bar, (b) Pressure drop at inlet pressure of 12 bar.

outlet air temperatures and pressures, as well as the mass flow rate from the testbench instrumentation.

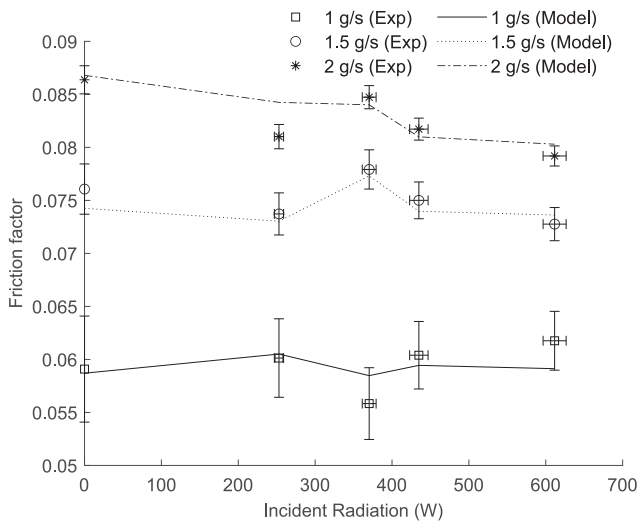
Observing Fig. 11, the friction factor remains, broadly, unchanging with increasing incident radiation at both 4 bar and 12 bar inlet pressures. Referring to Eq. (3), this behaviour can be understood as the increased channel velocity, due to higher average fluid temperatures at higher incident radiation levels, increases the pressure drop which counteracts the effect of this greater channel velocity and ultimately maintains the friction factor at a relatively stable level. On the other hand, the mass flow rate appears to have a small positive effect on the average friction factor though the values of average friction factor do not vary widely across mass flow rates with mean values of 0.073 and 0.0688 at 4 bar and 12 bar inlet pressures respectively at standard deviations of 0.011 and 0.014 at these respective pressure levels.

It should be noted that the numerical model of the absorbers [45], whose output is plotted along with the experimental results in Fig. 11, divides the absorber into multiple heat collector elements along the absorber length (flow direction) and calculates the friction factor and

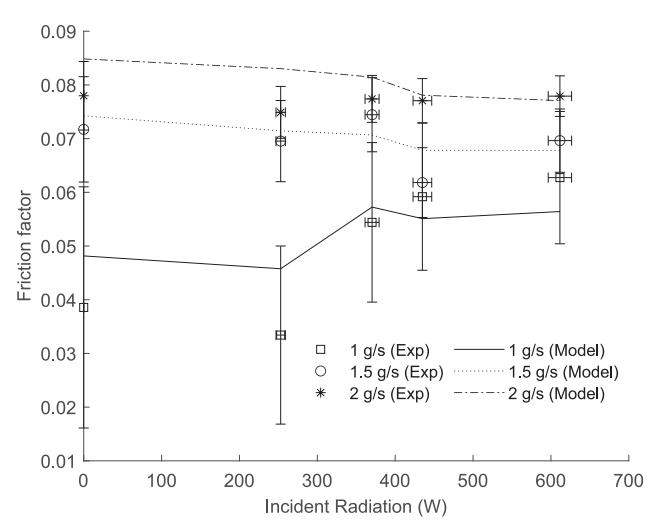
pressure drop sequentially from the absorber inlet to outlet. This is done as the air properties vary along the absorber length. It is the average friction factor, calculated after the overall pressure drop of the absorber is numerically determined, that is plotted in Fig. 11.

The variation of the outlet air temperature at different incident radiation flux levels and mass flow rates is depicted in Fig. 12. The lower and upper bands represent the results of the numerical model when fed with the absorptances before and after the experiment run respectively.

The outlet temperature is expected to rise with higher incident radiation values as the heat input to the HTF is greater. The mass flow rate is indirectly related to the outlet temperature as a lower flow rate must be compensated by a higher outlet temperature, assuming the heat transferred to the HTF is the same. Increased inlet pressures seem to slightly increase the outlet temperature as the effect of the higher density offsets the associated fall in channel flow velocity to improve the heat transfer to the fluid. Regarding the uncertainty related to the absorptances, it is observed in Fig. 12 that the outlet temperatures experimentally observed lie within the bounds of those predicted by the



(a)



(b)

Fig. 11. Experimental results of absorber PR1 compared with numerical model: (a) Friction factor at inlet pressure of 4 bar, (b) Friction factor at inlet pressure of 12 bar.

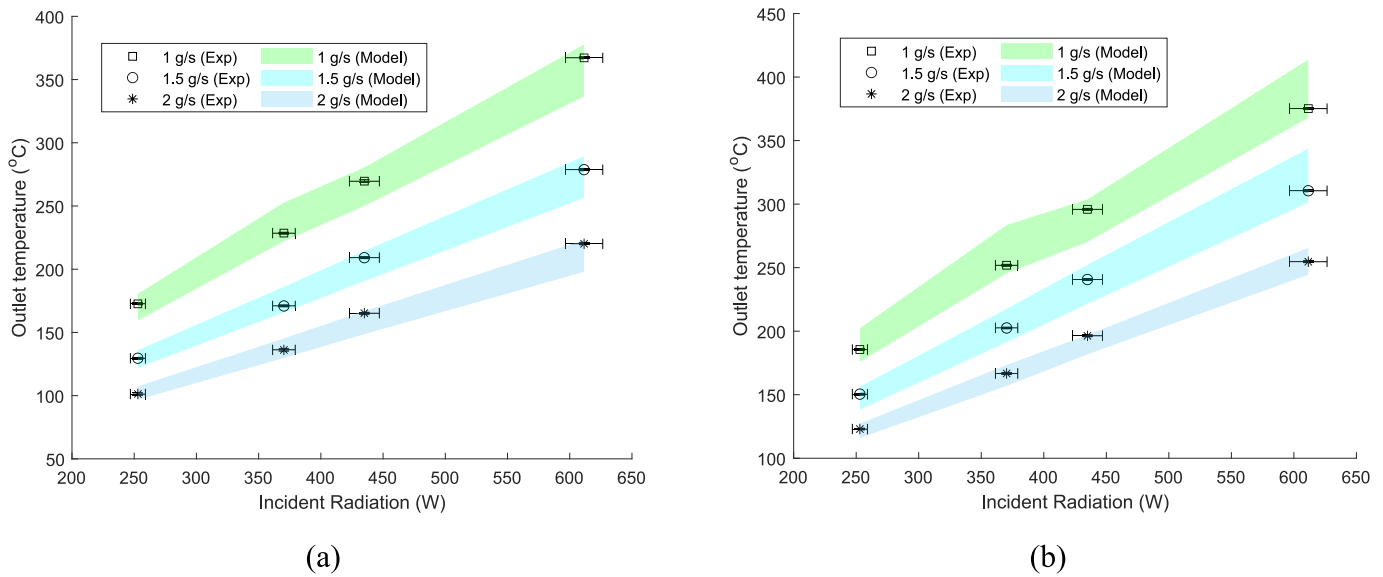


Fig. 12. Experimental results of PR1 absorber compared with numerical model. (a) Outlet temperature at inlet pressure of 4 bar, (b) Outlet temperature at inlet pressure of 12 bar.

numerical model when the absorptances before and after experimentation are fed to the model. In general, at lower radiation flux levels the outlet temperature is closer to the lower bound as significant oxidation has not occurred and the prevalent absorptance is closer to that measured before the experiment run. Besides, as the experiment run progresses, it is observed that the outlet temperatures at higher radiation flux levels are closer to the upper bounds. This is reasonable as at these elevated temperature conditions, extensive surface oxidation occurs and the absorptance subsequently rises to be more in line with the absorptance measured after the experiment run.

3.3. Comparative performance analysis of absorbers

From Table 6, it is evident that PR1 consistently has the highest pressure drop which is unsurprising as it has the smallest channels and hence the largest channel velocity which directly affects the pressure drop. As PR2 and PR3 have the same wetted area, the measured pressure drops are similar as the only differentiating influence on their pressure drop would be the heat conducted through the receiver bulk and hence transmitted to the fluid, which is a relatively weak effect. PR4 has the largest number of flow channels and thus, for the same mass flow rate, has the lowest fluid velocity and pressure drop. Of course, the surface roughness of each absorber may not be the same and that has an outsized effect on the pressure drop but this roughness is unknown and is assumed equal for the sake of this discussion.

It should be noted that most of the differences in measured pressure drop between absorbers, especially at higher input pressures and lower mass flow rates, are miniscule and cannot be concluded upon given the magnitude of measurement uncertainty. The omitted cells in Table 6 correspond to those operating points at which the measured pressure drop falls within the bounds of experimental error and hence these values are discarded.

At higher inlet pressures, the pressure drop is expectedly lower as the channel velocity is correspondingly lower due to the larger fluid density. The relative pressure drop is always below 1 % at 12 bar inlet pressures but rises to around 2.5 % when the inlet pressure is 4 bar. At its peak, for the PR1 absorber operating at 4 bar, peak incident radiation flux of 500 kW/m² and a mass flow rate of 2 g/s, the pressure drop is 7.5 %.

The energy and thermal efficiencies of the absorbers at the different operating conditions used in the experimental campaign are tabulated in Table 7. The efficiency values at some operating points are omitted

Table 6

Experimental absorber pressure drop.

Control parameter	Peak flux (kW/m ²)	Inlet pressure (bar)	Mass flow rate (g/s)	Pressure drop			
				PR1 (mbar)	PR2 (mbar)	PR3 (mbar)	PR4 (mbar)
0	4	1.0	42	19	—	14	
0	4	1.5	114	63	64	24	
0	4	2.0	244	72	69	96	
200	4	1.0	57	21	25	20	
200	4	1.5	136	69	33	33	
200	4	2.0	253	118	64	81	
300	4	1.0	58	31	25	15	
300	4	1.5	154	77	38	42	
300	4	2.0	277	124	79	78	
400	4	1.0	66	35	20	13	
400	4	1.5	158	79	45	45	
400	4	2.0	281	128	81	80	
500	4	1.0	78	0	0	0	
500	4	1.5	170	86	44	48	
500	4	2.0	299	138	80	96	
0	12	1.0	—	—	—	—	
0	12	1.5	47	11	0	6	
0	12	2.0	66	32	31	0	
200	12	1.0	9	7	7	12	
200	12	1.5	48	15	16	0	
200	12	2.0	84	36	30	22	
300	12	1.0	20	0	7	0	
300	12	1.5	56	21	13	7	
300	12	2.0	95	40	21	21	
400	12	1.0	23	—	—	—	
400	12	1.5	49	23	5	—	
400	12	2.0	101	38	22	20	
500	12	1.0	27	0	0	0	
500	12	1.5	61	20	6	7	
500	12	2.0	111	41	32	22	

because the experiment runs at those operating points were abandoned as the absorber solid temperature rose above safety limits defined in the experiment operating procedure. This overheating of the absorber primarily occurred at high incident radiation flux levels and low mass flow rates. As mentioned previously, the absorptances at each set of operating conditions changed as the previously unoxidised and otherwise untreated absorbers oxidise with the elevated surfaces temperatures. It was observed that the heaviest oxidation only occurred at higher fluxes and

Table 7

Experimental energy and thermal efficiency.

Control parameter			Energy Efficiency				Thermal Efficiency			
Peak flux (kW/m ²)	Inlet pressure (bar)	Mass flow rate (g/s)	PR1(%)	PR2(%)	PR3(%)	PR4(%)	PR1(%)	PR2(%)	PR3(%)	PR4(%)
200	4	1.0	59.9	64.2	62.9	65.5	75.6	67.2	54.1	56.4
200	4	1.5	62.6	67.2	66.0	71.0	84.8	88.7	60.3	73.5
200	4	2.0	61.8	67.8	68.4	76.2	95.8	94.2	90.5	91.6
300	4	1.0	56.2	57.5	56.3	60.9	71.0	60.2	48.4	52.4
300	4	1.5	60.1	62.8	61.7	67.3	81.4	82.8	56.4	69.6
300	4	2.0	61.6	64.5	64.6	72.0	95.5	91.4	88.1	90.9
400	4	1.0	57.5	58.2	59.6	62.8	72.6	60.9	51.3	54.0
400	4	1.5	64.6	63.7	63.0	67.9	83.6	79.9	57.6	70.3
400	4	2.0	66.0	66.8	66.2	71.4	94.3	90.2	90.2	90.2
500	4	1.0	58.1	—	—	—	73.4	—	—	—
500	4	1.5	63.6	64.7	62.2	67.6	78.4	81.2	56.9	69.9
500	4	2.0	65.6	66.6	64.6	70.6	93.7	89.9	88.1	89.2
200	12	1.0	66.5	—	69.7	66.9	72.1	78.9	50.3	55.0
200	12	1.5	78.0	72.9	70.6	75.3	81.1	84.7	55.3	72.5
200	12	2.0	79.1	73.8	75.7	79.1	93.4	93.2	68.8	76.8
300	12	1.0	64.3	65.8	62.7	62.3	69.7	70.6	45.2	51.2
300	12	1.5	75.2	66.9	65.3	71.7	78.2	77.7	51.1	69.1
300	12	2.0	78.8	70.3	70.4	76.3	93.1	90.5	64.0	74.1
400	12	1.0	65.3	65.1	62.9	63.6	70.7	69.8	45.4	52.2
400	12	1.5	77.4	67.5	68.2	72.2	80.5	74.8	53.4	69.5
400	12	2.0	82.8	72.9	72.4	78.9	90.0	89.4	65.8	76.7
500	12	1.0	60.2	—	—	—	65.2	—	—	—
500	12	1.5	72.8	65.6	63.6	69.0	75.7	72.6	49.8	66.4
500	12	2.0	79.2	71.1	68.2	74.9	86.1	87.2	62.0	72.8

low mass flow rates. More specifically at mass flow rates of 1 g/s, and to a lesser extent 1.5 g/s, and incident radiation peak fluxes of 400–500 kW/m², the absorbers underwent severe oxidation which altered all subsequent experiments on the same absorber. At elevated temperatures, stainless steel 316L develops a protective oxide layer, of principally chromium oxides [61,62], that has a darker colour and an increased absorptance [63,64]. The increased absorptance causes more heat to be absorbed from the incident radiation which, in turn, leads to added heat transfer to the gas. As more radiation power is transferred to the gas, the energy efficiency of the absorber improves. The higher temperature of the gas causes also increases its specific volume which results in a greater fluid velocity and higher pressure drops. This complements the heat transfer as increased channel velocity and turbulence, due to the greater pressure drop, enhances the convective heat transfer.

As the experiment plan called for experiments at the lower pressure level (4 bar) to be conducted first, the oxidation on the absorber, and hence its absorptance, during the 12 bar experiment runs were greater leading to better energy efficiencies at higher pressures in Table 7 for all absorbers. The lack of consideration of a varying absorptance makes energy efficiency a less reliable performance indicator than the thermal efficiency in this instance. The uncertainties of the energy and thermal efficiencies were challenging to ascertain as multiple experimental variables, each with their own uncertainties, interacted and combined into more complex terms [65]. An error analysis has been performed conforming to the International Organization for Standardization (ISO) guide [66] which employs the Taylor Series Method (TSM) to model experiment errors and their propagation. It revealed that the maximum and mean absolute error in the energy efficiency was 2.7 % and 2.15 % respectively and the maximum and mean absolute error in the thermal efficiency was 3.6 % and 2.8 % respectively. A sensitivity analysis was performed [51] to gain further insight into the interactions and influences that shape the efficiencies but it is considered beyond the scope of this work.

Regarding the thermal efficiency, multiple interesting trends are observed. It falls with increasing radiation flux which is expected given the higher thermal losses at the elevated surface temperatures corresponding to higher incident radiation. The mass flow rate has a significant impact on the thermal efficiency with greater mass flow rates leading to increased efficiencies due to enhanced turbulence-led heat

transfer and heat evacuation from the absorber surface owing to reduced thermal resistance in the absorber bulk. The inlet pressure has a less noticeable effect though it is generally observed that thermal efficiencies were slightly higher at lower pressure levels which is explained by increased channel flow velocities, and associated heat transfer coefficients, at lower pressures.

The highest thermal efficiency is observed in PR1 at 95.8 % though it also performs better than the other absorbers at most other operation points. This improved efficiency is expected given that PR1 has the smallest channel size and wetted area and thus the largest channel flow velocity which ultimately results in the best heat transfer to fluid. PR2, however, follows closely behind PR1 in terms of thermal efficiency which verifies the findings of the numerical model which found improved thermal performance with taller and narrower channels with thicker walls due to improved thermal conduction through the solid walls. In line with this understanding, PR4 has worse thermal efficiency than PR2 due to thinner channel walls and the additional horizontal flow channel. Finally, PR3 exhibits the worst thermal efficiency which was expected as the broader flow channels increased the thermal resistance of the absorber which ultimately resulted in higher thermal losses and worse heat transfer to fluid.

The pressure drop at the highest thermal efficiency (PR1) is 6.3 % of the inlet pressure and similar values of relative pressure drop are observed at thermal efficiencies above 90 % at inlet pressures of 4 bar though these fall to around 0.75 % at inlet pressures of 12 bar. Though the thermal efficiencies of PR2 are slightly lower than those corresponding to PR1, the pressure drops are also lower with relative pressure drops around 3 % of the inlet pressure at 4 bar. Again, the relative pressure drops for PR2 crash to just 0.33 % at inlet pressures of 12 bar. This not only experimentally demonstrates that the absorbers achieve a high thermal efficiency without excessive pressure drop but also indicates that, as predicted by the numerical model, an optimum absorber geometrical configuration has taller and narrower channels.

4. Conclusions and future work

This study aimed to experimentally validate a previously conducted numerical analysis on the influence of internal flow channel geometries on the performance of solar absorbers. The primary goal was to confirm

the predictions of the numerical model regarding optimal channel geometries that maximise heat transfer from solar to fluid while minimising pressure drop.

The experimental setup included a pressurised receiver test facility, utilising the KIRAN-42 High Flux Solar Simulator as the concentrated light source. Four stainless steel absorbers, designed with varying rectangular channel aspect ratios and wall thicknesses, were manufactured using Selective Laser Melting (SLM) for this investigation. The experimentation involved characterising each absorber by varying the incident radiation flux, mass flow rate, and inlet pressure to observe their performance.

The experimental results indicated that the absorber with the smallest channels achieved the highest thermal efficiency of 95.8 %, surpassing several state-of-the-art receivers that typically do not exceed 90 % efficiency. Furthermore, this absorber demonstrated a low pressure drop of 6.3 % of the inlet pressure, which is superior to similar state-of-the-art receivers operating at comparable thermal efficiencies. Other absorbers, which had increased channel height, showed slightly lower thermal efficiencies, although still above 90 %, and even lower relative pressure drops of less than 3 %.

The experiment campaign was conducted using pressurised air as the heat transfer fluid and the thermal properties of the fluid, including its density, thermal conductivity, specific heat etc, have an impact on the absorber performance. The results of this study should thus be considered as specific to pressurised air. Compact channel solar absorbers may employ a multitude of heat transfer fluids that, for the same flow channel geometry, may exhibit better or worse performance as has been observed in this experimental work.

These findings confirmed the numerical model's predictions, showing that absorbers with smaller channels provide better heat transfer to the fluid, and that optimal flow channel geometries might feature taller and narrower channels. The study also examined the impact of channel wall thickness, but its influence was less clear compared to other parameters. A noticeable decline in performance was

observed in the absorber with increased channel breadth, aligning with the numerical model's expectations.

One significant source of uncertainty in the experimental results was the absorptance of the absorbers. The stainless steel absorbers were untreated and unoxidized, leading to changes in absorptance as the temperature of the absorber solid increased during the experiments. The absorptance was measured before and after the experiments using a spectrophotometer and image processing, which introduced variability in the results. This issue could have been mitigated by either measuring the absorptance after each experiment run or by pre-treating the absorbers to ensure a more consistent absorptance, with the latter being preferable for simplicity and time efficiency.

Further experimental work can be carried out using the developed pressurised receiver test facility. Future studies could involve evaluating absorbers made from different materials or with various surface treatments under different conditions, such as varying mass flow rates, inlet pressures, working gases, and incident radiation levels. Additionally, receiver sizes and configurations could also be tested within the facility's limits.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

A.1 Modifications to pressurised gas receiver numerical model

In this section the modifications made to original numerical model [51], which was modified to consider several experimental items that were not initially taken into account, are briefly described below.

(1) Inhomogeneous incident radiation flux distribution

The original numerical model assumed a uniform incident radiation flux distribution across the absorber which is not the case in this experimental. As the numerical model divides the receiver into heat collector elements (HCEs) along the flow direction/absorber length, the model was slightly modified to allow for the input of different incident radiation power for each HCE. The incident radiation flux on the receiver has been integrated along the receiver breadth for each HCE (i.e. pixel along the absorber breadth in the flux map image) to determine the incident radiation on each HCE. The number of HCEs depends on the pixel matrix that comprises the flux map image and more specifically the number of pixels in the direction along the absorber length. As the number of HCEs is a controllable input in the model, the incident radiation per HCE obtained from the flux map processing can be easily fed into the absorber model. The processing of the flux maps into incident radiation distributions along the absorber length is shown in Fig. A1.

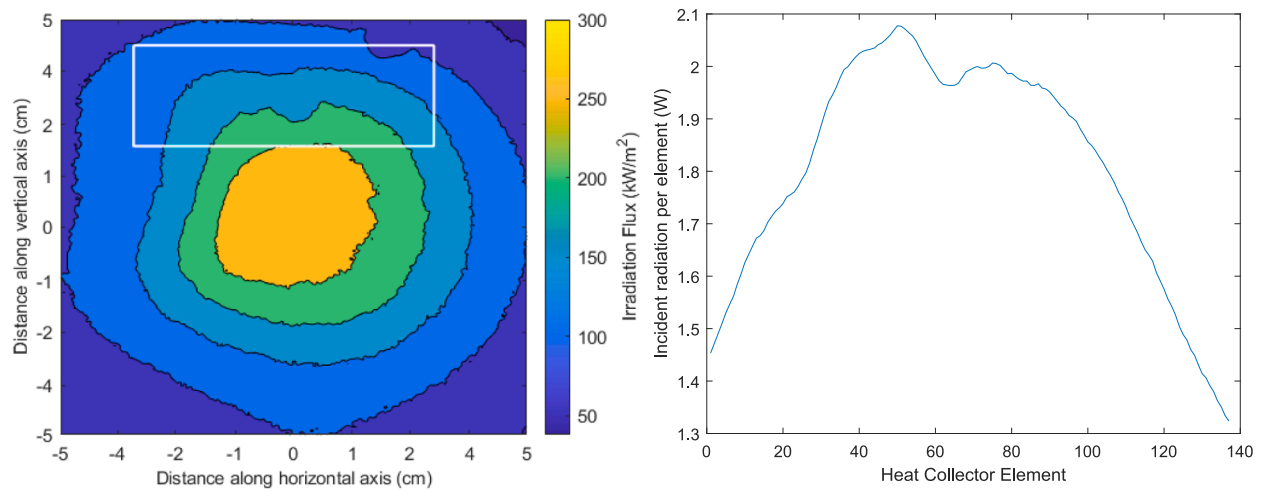


Fig. A1. Irradiance map processing to determine incident radiation on each HCE for the different peak flux levels used in the experimental campaign. (Left) Flux map with receiver marked in white rectangle, and (right) incident radiation per HCE from absorber inlet to outlet. (a, b) 200 kW/m², (c, d) 300 kW/m²; (e, f) 400 kW/m²; and (g, h) 500 kW/m².

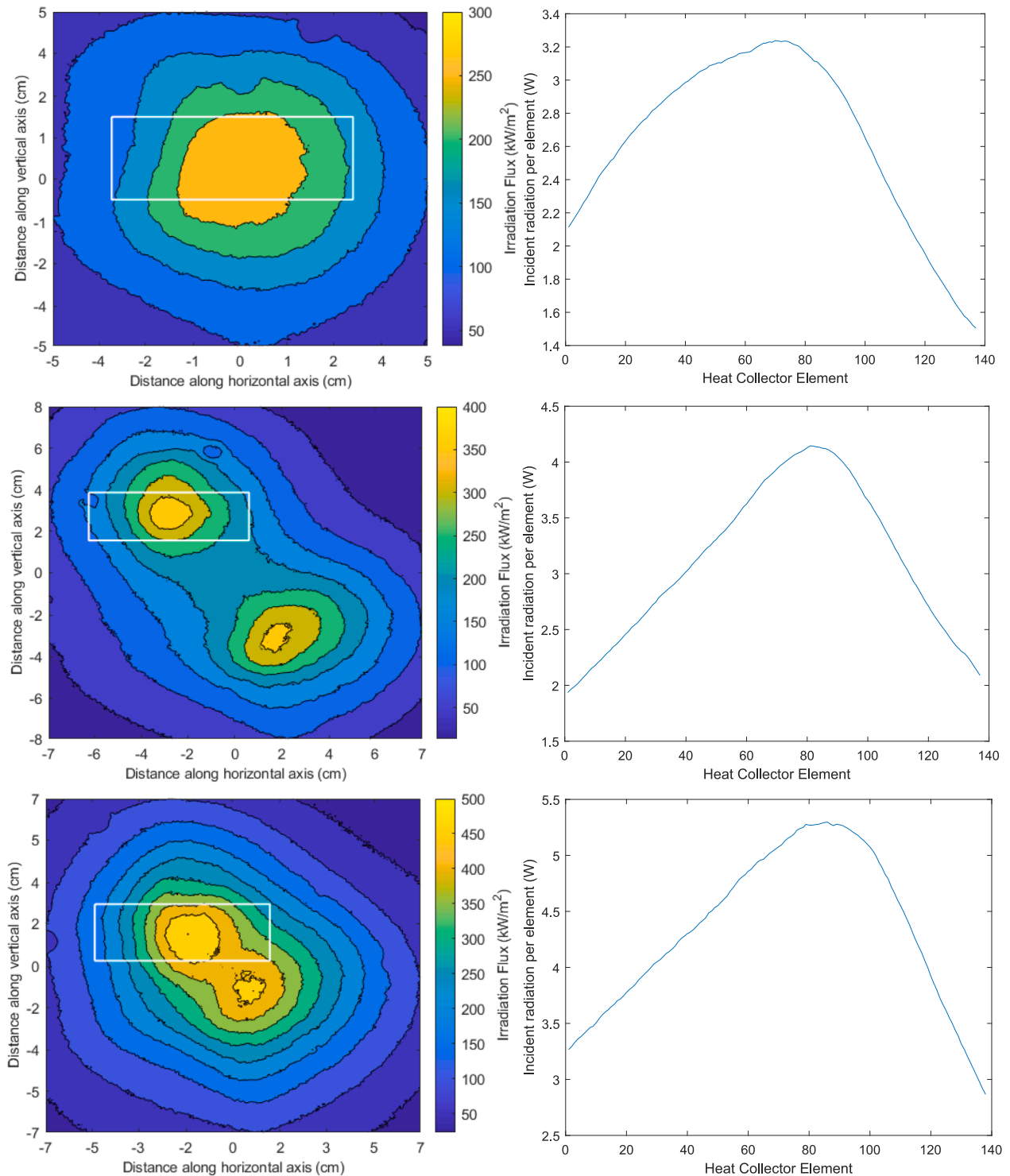


Fig. A1. (continued).

(2) Roughness of flow channel walls

The original numerical model facilitates the use of whichever empirical or semi-empirical correlation best suits the situation at hand. In the case of the experimental absorbers, it was clear from the off that the ‘smooth channel’ assumption employed earlier would not be suitable as surface roughness and imperfections due to additive manufacturing is a well-documented phenomenon [67,68]. Previous studies on the roughness of AISI 316L steel pieces produced by 3D printing reported roughness values (ϵ) of around 15 μm [59,60].

The addition of channel wall roughness to the model increases the pressure drop of the HTF due to increased friction with the channel but this increased friction and turbulence also results in better heat transfer to the HTF [69]. New correlations for the pressure drop and heat transfer modelling must be implemented in the numerical model to account for the surface roughness.

The pressure drop correlation used is the Nikuradse equation [70] which is given in Eq. (4);

$$\frac{1}{\sqrt{f_D}} = 3.48 - 1.737 \log\left(\frac{\varepsilon}{r_h}\right), \quad (4)$$

where f_D is the Darcy friction factor, ε is the surface roughness value (Ra) and r_h is the hydraulic radius.

The heat transfer correlation used is the Martinelli equation [71] which is expressed in Eq. (5);

$$Nu = \frac{Re Pr \sqrt{f_F/2}}{5 \left[Pr + \log(1 + 5 Pr) + 0.5 \log\left(\frac{Re \sqrt{f_F/2}}{60}\right) \right]}, \quad (5)$$

where Nu is the Nusselt number, Pr is the Prandtl number, Re is the Reynolds number and f_F is the Fanning friction factor.

(3) Time-dependent absorptivity

At elevated temperatures, an oxide layer develops on absorber surface which drastically alters its absorptivity. To quantify this, absorption measurements were conducted on the inlet, central and outlet sections of the absorber before starting after concluding the experiments. Hemispherical absorbance measurements were performed using a spectrophotometer UV-Vis-IR Lambda 1050 and integrating sphere 150 mm InGaAs (Perkin Elmer). The result of such a measurement on an unoxidised absorber sample, tested as received from the supplier i.e. before any high temperature experiment had been performed on it, can be seen in Fig. A2.

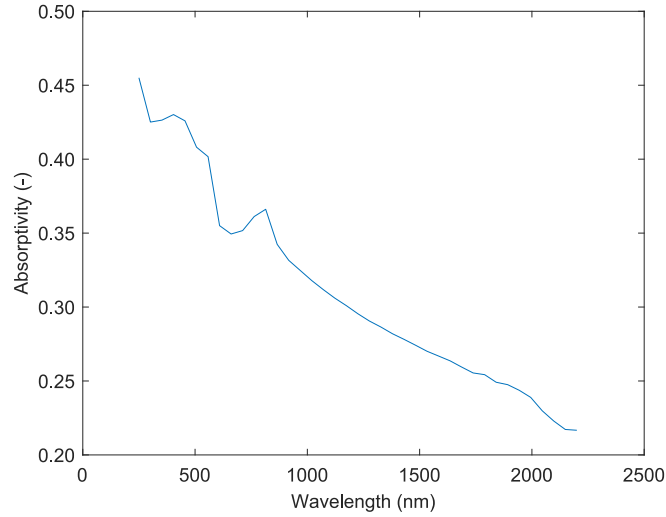


Fig. A2. Absorptivity of unoxidised receiver sample as received from supplier.

Combining the absorptivity measurements with the xenon arc lamp emission spectrum, shown in Fig. A3.(a), results in the absorbed radiation profile plotted in Fig. A3.(b). The term absorptance, henceforth, refers to the weighted Xenon arc spectrum absorptance. Three repetitions of spectroscopy tests were performed resulting in three absorptance values the average of which is chosen as the final value. The results of these tests are given in Table A.1.

Table A1
Absorptance of unoxidised absorber.

Test No.	Absorptance (-)
1	0.2631
2	0.2602
3	0.2592
Average absorptance	0.2608
Standard deviation	0.0020

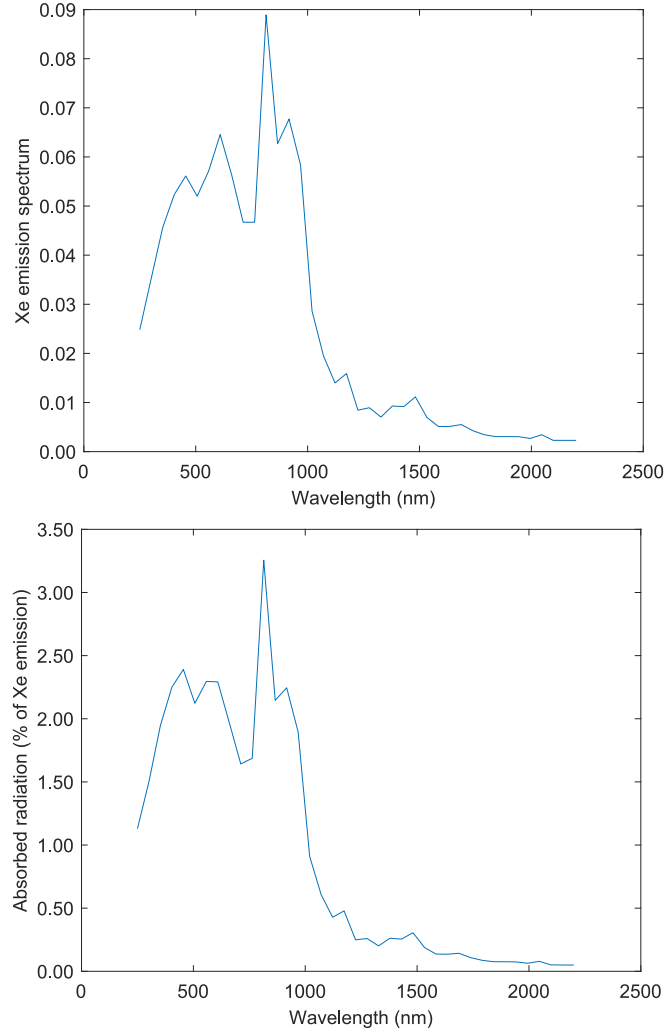


Fig. A3. Absorbed radiation analysis: (top) Xenon emission spectrum, (bottom) Absorbed radiation

After this absorptance is measured, it is multiplied with the incident radiation on the absorber to determine the absorbed radiation.

(4) Additional pressure drops

Referring to the pressurised receiver testbed layout in Fig. 2, there are additional sources of loss of pressure in the system, in addition to the absorber itself, from the inlet pressure sensor to the outlet pressure sensor. These are the pipe sections and sudden contraction and expansion zones at the absorber inlet and outlet respectively which can be estimated [72] by Eq. (6):

$$\Delta p = \Delta p_{pipe} + \Delta p_c + \Delta p_e = \frac{1}{2} \rho v^2 \left(f_D \frac{L_{pipe}}{D_{hpipe}} + K_c + K_e \right), \quad (6)$$

where Δp_{pipe} refers to the pressure drop in the pipe section; Δp_c and Δp_e are the pressure drop due to contraction and expansion, respectively; L_{pipe} refers to the length of the pipe section; and K_c and K_e are the contraction and expansion coefficients [72]. While most variables in Eq. (5) varied with operating conditions, the geometric parameters of the pipe length between absorber inlet/outlet and pressure sensor and hydraulic diameter were constant at 20 cm and 10 mm respectively.

A.2 General experiment operating procedure

The experimentation was performed following the operating procedure shown in Fig. A4, after the precautionary and preparatory measures have been done as detailed in [51].

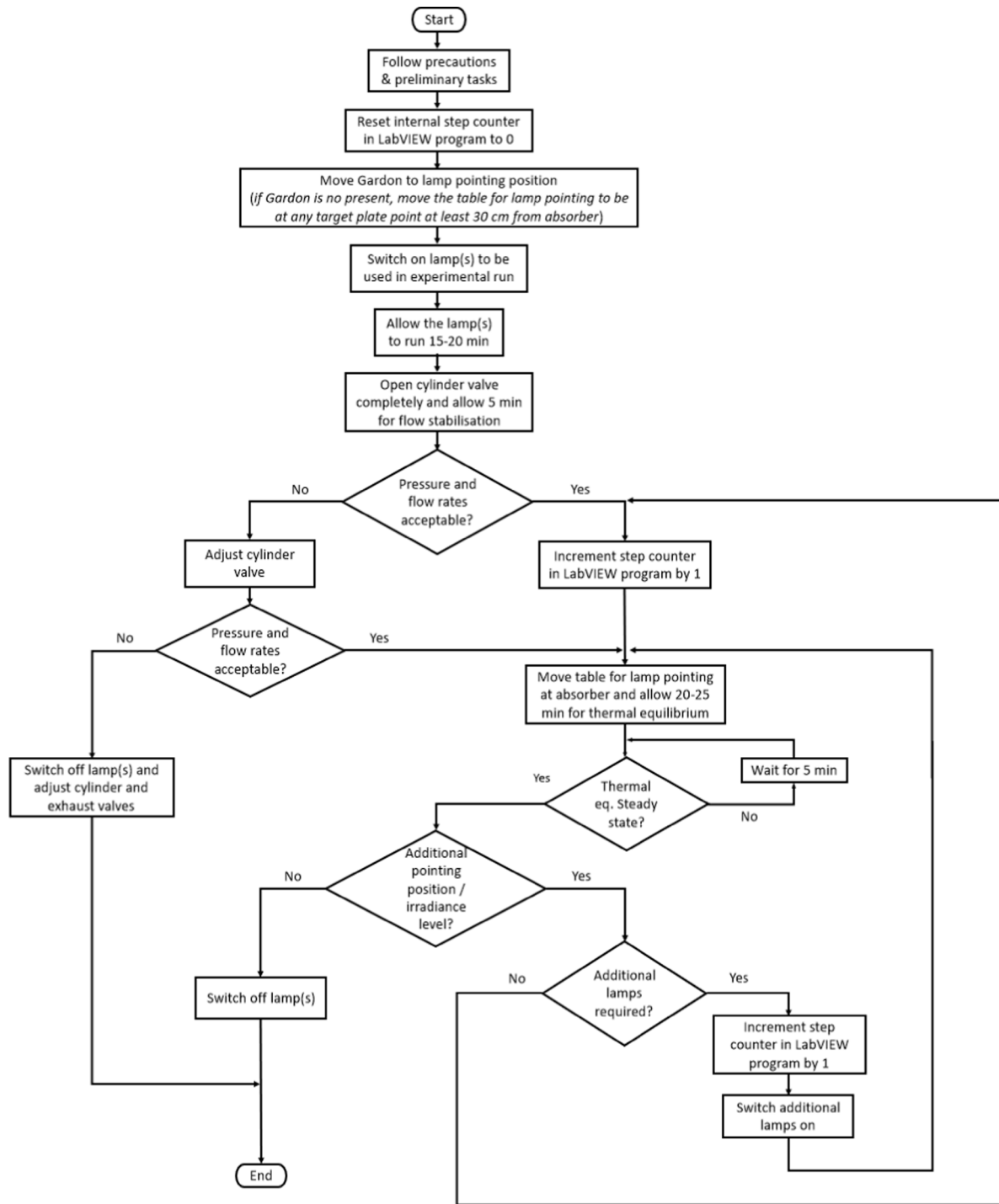


Fig. A4. Experiment procedure for normal experiment run on pressurised receiver testbed.

Data availability

Additional data has been shared in [Appendix](#). Furthermore, there is a reference to PhD thesis which contains additional data on experimental and numerical work besides more elaborate explanation.

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